

Status	Finished
Started	Monday, 3 November 2025, 10:24 AM
Completed	Monday, 3 November 2025, 10:48 AM
Duration	24 mins 33 secs

Question **1**

Correct

The k-digit number N is an Armstrong number if and only if the k-th power of each digit sums to N.

Given a positive integer N, return true if and only if it is an Armstrong number.

Example 1:

Input:

153

Output:

true

Explanation:

153 is a 3-digit number, and $153 = 1^3 + 5^3 + 3^3$.

Example 2:

Input:

123

Output:

false

Explanation:

123 is a 3-digit number, and $123 \neq 1^3 + 2^3 + 3^3 = 36$.

Example 3:

Input:

1634

Output:

true

Note:

$1 \leq N \leq 10^8$

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  #include<math.h>
3
4  int main(){
5      int n,temp,rem,digits=0;
6      long sum=0;
7      scanf("%d", &n);
8      temp=n;
9
10     while(temp){
11         digits++;
12         temp/=10;
13     }
14     temp=n;
15     while(temp){
16         rem=temp%10;
17         sum+=pow(rem,digits);
18         temp/=10;
19     }
20     printf(sum==n ?"true":"false");
21
22
23 }
24
```



	Input	Expected	Got	
✓	153	true	true	✓
✓	123	false	false	✓

Passed all tests! ✓

Question **2**

Correct

Take a number, reverse it and add it to the original number until the obtained number is a palindrome.

Constraints $1 \leq \text{num} \leq 999999999$ **Sample Input 1**

32

Sample Output 1

55

For example:

Input	Result
32	55
1234	5555

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  long rev(long n){
3      long r=0;
4      while(n>0){
5          r=r*10+n%10;
6          n/=10;
7      }
8      return r;
9  }
10 }
11 int main(){
12     long n;
13     scanf("%ld", &n);
14     while(n!=rev(n))
15         n+=rev(n);
16     printf("%ld",n);
17 }
```



	Input	Expected	Got	
✓	32	55	55	✓
✓	1234	5555	5555	✓

Passed all tests! ✓



Question **3**

Correct

Maya, a student in an arts and crafts class, wants to create a pattern using stars (*) in a specific format. She plans to use a program to help her construct the pattern.

Write a program that takes an integer as input and constructs the following pattern using nested for loops.

Input: 5

Output:

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
*
```

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main(){
3      int n,i,j;
4      scanf("%d",&n);
5      for(i=1; i<=n;i++){
6          for(j=1; j<=i;j++) printf("* ");
7          printf("\n");
8          for(i=n-1; i>=1;i--){
9              for(j=1;j<=i;j++) printf("* ");
10
11              printf("\n");
12          }
13 }
```



	Input	Expected	Got	
✓	5	<pre>* *</pre>	<pre>* *</pre>	✓

Passed all tests! ✓